PREDICTING NEXT-YEAR HEALTH CARE COSTS AND UTILIZATION USING MEPS SURVEY DATA

Machine-learning risk ranking for high cost, emergency department use, and inpatient events with age- and sex-based subgroup evaluation

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1. Executive Summary

Healthcare systems and insurers need to identify individuals at risk of **high future costs** and **acute events** (emergency department (ED) visits and inpatient stays) to allocate limited care management resources efficiently. This thesis develops and evaluates person-level prediction models using the **MEPS Household Component, Panel 27 (2022–2023; HC-252)**.

Using **Year-1** information to predict **Year-2** outcomes, we study four targets: (i)Year-2 total expenditures(log-transformed), (ii) Year-2 high-cost status, (iii) any ED visit in Year-2, and (iv) any inpatient stay in Year-2. We compare baseline regression models with machine-learning approaches and evaluate performance using ranking-focused metrics suitable for imbalanced outcomes, complemented by subgroup robustness checks.

Overall, results support using the models better suited for ranking/selection than predicting exact outcomes for each individual. The predictive signal is strongest for **high-cost status**, moderate for **ED risk**, and weakest for **inpatient events**, likely because inpatient events are often driven by acute shocks not captured in Year-1 survey data. Across the four targets (log-cost, high-cost, ED, IP), **prior-year spending and utilization** are the most informative predictors, with additional contributions from age, insurance and baseline health indicators. Finally, we present a lightweight MVP ("Health Outcome Predictor") illustrating how top-k selection and clear outputs can translate model results into an actionable decision workflow.

2. Problem Statement & Hypotheses

2.1 Problem Statement

Healthcare spending in the United States is highly concentrated: a very small share of individuals accounts for a disproportionately large fraction of total expenditures. Recent analyses using the Medical Expenditure Panel Survey (MEPS) Household Component show that in 2022 the top 5% of spenders accounted for roughly half of all healthcare expenditures, while the bottom 50% accounted for less than 3%. (Hernandez-Viver & Mitchell, 2025) Identifying such high-cost individuals prospectively and understanding the drivers of their spending is a central challenge for healthcare systems and insurers.

The **Medical Expenditure Panel Survey – Household Component (MEPS-HC)** is a nationally representative longitudinal survey of the U.S. civilian non-institutionalized population, following each panel for five interviews over two calendar years and collecting detailed information on health status, medical conditions, utilization, expenditures and insurance coverage. (Cohen, 1997) Panel 27 (2022–2023), released as the longitudinal file **HC-252**, links person-level information across two full years and is therefore well suited to studying how Year-1 characteristics predict Year-2 outcomes.

Prior research using MEPS has demonstrated that information on chronic conditions, self-reported health and functional limitations substantially improves prediction of future high expenditures compared with models using only age and gender. (Fleishman & Cohen, 2010) Beyond clinical factors, social determinants of health and socio-economic status have been shown to be strongly associated with health care expenditures and to vary in their impact across insurance types. (Mohan & Gaskin, 2024) A large literature also documents that ED visits and hospitalizations are concentrated among patients with specific demographic, socio-economic and clinical profiles, suggesting that these events may be predictable from prior information. (Sun et al., 2003)

At the same time, there is growing evidence that prediction algorithms in healthcare can systematically under-serve certain groups. For example, a widely used commercial risk-stratification algorithm that relied on past costs as a proxy for health needs was shown to substantially underestimate the needs of black patients compared with white patients at the same risk score. (Obermeyer et al., 2019) More recent work has explicitly evaluated fairness metrics for machine-learning models predicting hospitalization and ED visits, finding sizeable performance gaps across demographic and socio-economic subgroups. (Davoudi et al., 2024)

Against this background, the central problems addressed in this thesis are:

1. **Prospective prediction of total expenditures.**
   How accurately can Year-2 total health care expenditures be predicted from information available at the end of Year-1, including socio-economic status (SES), insurance coverage, baseline health and prior utilization?
2. **Identification of high-cost individuals.**
   Can we reliably classify individuals into high-cost versus non-high-cost groups for Year-2, and which predictors are most important for this classification?
3. **Prediction of acute events.**
   To what extent can Year-2 emergency room visits, and inpatient stays be predicted using Year-1 characteristics and prior utilization patterns?
4. **Equity and subgroup performance.**
   Do model performance and prediction errors differ systematically across subgroups defined by age, sex, and what are the implications for equity and potential bias?
5. **Practical use and productization.**
   How could a prediction model based on MEPS variables be translated into a simple, explainable tool that could support decision-making in practice, for example in an insurer or care-management context?

2.2 Research Questions and Hypotheses

Building on prior work on high-cost patients, predictors of future expenditures and acute utilization, and emerging evidence on fairness in health-care prediction models, this thesis formulates the following research questions (RQ) and Hypotheses (H):

RQ1. Which individual-level factors are most strongly associated with Year-2 total health care expenditures?

- **H1.** Higher baseline expenditures, a greater burden of chronic conditions and poorer self-reported health are associated with higher Year-2 total expenditures, even after adjusting for socio-economic and insurance covariates. (Fleishman & Cohen, 2010; Mohan & Gaskin, 2024)

RQ2. How well can we distinguish high-cost individuals in Year-2 using Year-1 information?

- **H2.** Models that combine socio-demographic variables, poverty status, insurance type, chronic conditions and prior utilization will achieve substantially better discrimination of high-cost individuals than simple baselines based only on age and prior spending. (Fleishman & Cohen, 2010)

RQ3. Which factors predict Year-2 emergency room visits and inpatient stays?

- **H3.** Lack of insurance or gaps in coverage, lower socio-economic status and higher baseline use of emergency and inpatient services are associated with increased risk of Year-2 ED visits and hospitalizations. (Sun et al., 2003; Mohan & Gaskin, 2024)

RQ4. Is model performance consistent across socio-demographic subgroups?

- **H4.** Prediction errors and discrimination metrics differ across subgroups defined by income, race/ethnicity and insurance type, reflecting underlying structural differences in access to care and utilization patterns. These differences may raise fairness concerns that need to be explicitly assessed and discussed. (Obermeyer et al., 2019; Davoudi et al., 2024)

These research questions and hypotheses guide the design of the empirical analysis, the choice of variables and outcomes, and the evaluation strategy for the prediction models developed in subsequent chapters.

3. Data & Sampling Design

3.1 The Medical Expenditure Panel Survey

The empirical analysis is based on the **Medical Expenditure Panel Survey – Household Component (MEPS-HC)**, conducted by the U.S. Agency for Healthcare Research and Quality (AHRQ) and the National Center for Health Statistics.

MEPS uses an overlapping panel design: each household panel is followed for five interview rounds over approximately 2 years, covering two full calendar years of data. In any given year, information is collected from two panels: one in its first year of data collection and one in its second. Annual full-year consolidated files summaries person-level information for each calendar year, while separate longitudinal files link the two years for each panel.

Detailed information on the design and data-collection procedures of the MEPS Household Component, including the construction of person-level weights and the variance-estimation strata and primary sampling units—is provided in Cohen's (1997) methodology report *Design and Methods of the Medical Expenditure Panel Survey Household Component*.

3.2 Panel 27 and the HC-252 Longitudinal File

This thesis uses the **Panel 27 Longitudinal Data File, HC-252**, which links the 2022 and 2023 full-year consolidated files **HC-243** and **HC-251** at the individual level. HC-252 contains one record per sampled individual and provides:

- survey administration and panel participation variables.
- demographic and socio-economic characteristics.
- health insurance coverage and employment variables.
- self-reported health status and priority health conditions.
- annual utilization and expenditure summaries for 2022 (Year-1) and 2023 (Year-2).
- person-level longitudinal weights (**LONGWT**) and variance estimation variables (**VARSTR, VARPSU**). VARSTR and VARPSU are survey design variables used for variance estimation under complex sampling.

The public-use HC-252 file includes **8,292 individuals** and **2,648 variables**. For the present analysis, the official Excel version of HC-252 (h252.xlsx) was imported into Python and used as the raw dataset.

3.3 Initial Sample and Panel Participation

At the current stage of the work, no sample restriction has yet been applied in the data file: all 8,292 records in HC-252 are retained in the raw DataFrame (df_raw). Several variables, however, are specifically designed to support the construction of appropriate analytic samples for longitudinal analysis:

- **DUPERSID**: a unique person identifier that can be used to link across MEPS files.
- **PANEL**: the panel number (27 for all observations in HC-252).
- **YEARIND**: indicator of whether an individual is in scope for both years or only one calendar year.
- **ALL5RDS**: indicator that the person is in scope and has data in all five interview rounds.
- **DIED, INST, MILITARY, ENTRSRVY, LEFTUS, OTHER**: variables describing death during the panel, institutionalization, active-duty military status, late entry into the survey, leaving the U.S., and other special situations.

In later stages of the thesis, these variables will be used to define a main analytic sample that represents individuals in scope for both calendar years and with complete longitudinal information (for example, restricting to YEARIND=1 and ALL5RDS=1).

For survey-weighted analysis, HC-252 provides a longitudinal person weight **LONGWT**, which allows estimates to be generalized to the U.S. civilian non-institutionalized population for the 2022–2023 period, and design variables **VARSTR** and **VARPSU** for variance estimation under the complex sample design.

3.4 Selection of Core Variables

Given the very high dimensionality of the raw HC-252 file (2,648 variables), we concluded a key early design decision was to define a manageable and conceptually coherent set of **core variables** to be used for exploratory data analysis and modelling. Using the official codebooks and documentation for HC-252, HC-243 and HC-251, we systematically reviewed variable labels and the variable-source crosswalks to identify variables in the following domains:

- identification and survey design.
- demographics (age, sex, race/ethnicity, region).
- socio-economic status and family structure (family income, poverty status, household size).
- health insurance coverage and employment.
- self-reported general and mental health.
- priority chronic conditions (e.g. hypertension, diabetes, coronary heart disease, stroke, asthma, high cholesterol).
- year-specific utilization and expenditures, including total health care expenditures, emergency room visits, inpatient discharges, and expenditure by payer type.

These variables were assembled manually into a white-list selected based on literature relevance, Year-1 availability, missingness, and interpretability, and implemented in a Python function select_core_columns() within the project's preprocessing module. Applying this function to the raw dataset produces a reduced DataFrame (df_core) that retains all **8,292 individuals** but only **103 variables**, corresponding to the chosen core set.

4. Phase I – Exploratory Data Analysis

4.1 Light preprocessing

Before descriptive analysis, a minimal cleaning step was applied to the consolidated dataset: (i) MEPS special missing codes were recoded to standard missing values (NaN); (ii) any remaining logically invalid negative values in expenditure/payment variables were set to missing; and (iii) the sample was restricted to individuals observed in both calendar years with all five rounds (YEARIND=1, ALL5RDS=1). After these steps, the analytic sample includes **7,812 individuals**. No outlier removal or imputation was performed at this stage; these are addressed in Phase II for model-ready feature construction.

4.2 Overview of predictors (high-level)

The dataset covers the full life course with a near-balanced sex distribution, and includes substantial socio-economic heterogeneity (income is strongly right-skewed and poverty categories are used for subgroup stratification in descriptive analysis). Insurance coverage is relatively stable across years, with private and public coverage representing the majority of individuals. Health status measures show that most respondents report good-to-excellent health, while chronic conditions (e.g., hypertension, diabetes, asthma) are prevalent enough to provide meaningful clinical signals. Detailed distributions for demographics/SES, insurance/employment, and health variables are provided in **Appendix A (Figures A.1–A.7)**.

4.3 Missing data patterns

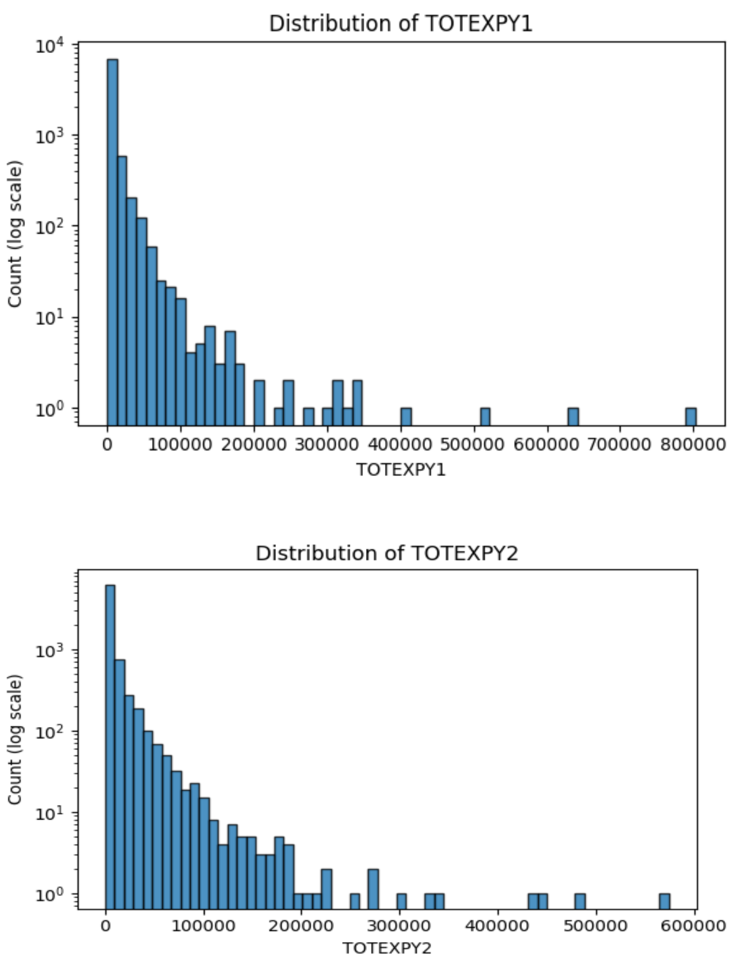
At the group level, average missingness is lowest for demographics, SES, insurance coverage and utilization/cost variables; higher missingness is mainly driven by **inapplicability/skip patterns**, especially for employment-related variables and narrow coverage-history flags; moderate for chronic conditions. Variables that are missing for the vast majority of the sample (e.g., PREVCOVR, MORECOVR) are excluded from the modelling feature set. This pattern suggests that a complete-case approach is acceptable for core demographic and SES variables, while more careful treatment is needed for employment and clinical indicators.

4.4 Outcomes and utilization (modelling-relevant findings)

Year-2 total expenditure (TOTEXPY2) exhibits a highly right-skewed, heavy-tailed distribution, with a small minority accounting for extremely high costs. This motivates modelling expenditures on the log scale using **log1p(TOTEXPY2)**. Figures **4.4-1–4.4-2** illustrate the heavy-tail pattern. Acute care outcomes are also strongly imbalanced: approximately **14%** of individuals have any ED visit in Year-2 (ANY_ED_Y2=1) and about **7%** have any inpatient stay (ANY_IP_Y2=1). Count variables for ED and inpatient use are highly concentrated at zero, motivating binary event modelling as the primary approach. Payer-specific expenditure distributions are presented in **Appendix A (Figure A.8)**.

Overall, the EDA confirms the expected MEPS patterns—skewed expenditures, concentrated acute utilization, and broad socio-economic/clinical heterogeneity. And this directly informs the modelling strategy in Phase II–III (log-transform for costs; binary outcomes and ranking metrics for acute events).

Figure 4. 4-1 Distribution of TOTEXPY1 / Figure 4.4-2: Distribution of TOTEXPY2



5. Phase II: Data Preparation & Feature Engineering

5.1 Preprocessing and analytic dataset

Phase I described the main distributional properties of the MEPS Panel 27 core dataset (e.g., strongly right-skewed expenditures and relatively low prevalence of ED visits and inpatient stays). In Phase II, the goal is to create a modelling-ready table by applying a reproducible preprocessing pipeline and constructing targets and predictors.

We start from the curated core-variable dataset and restrict the analytic sample to individuals in scope for both years with complete panel participation (**YEARIND=1, ALL5RDS=1**), yielding **N=7,812** observations. Special MEPS missing codes (e.g., inapplicable/refused/don't know) are recoded to standard missing values (NaN). Negative values in MEPS are typically special codes for missing/inapplicable responses; any remaining negative expenditure amounts are treated as invalid artifacts (e.g., from data adjustments) and recoded to missing. Variables with extremely high missingness and limited modelling value (e.g., coverage-history flags) are excluded from feature construction.

The resulting cleaned dataset (df_pre) is then passed into the feature engineering pipeline to produce df_feat.

5.2 Target construction

Targets are created from Year-2 outcomes:

- **Regression target**

$$LOG\_TOTEXPY2 = log\ (1 + TOTEXPY2)$$

which reduces the heavy-tailed skewness of annual expenditures.

- **High-cost classification target**

$$HIGHCOST\_Y2 = 1(TOTEXPY2 \geq q_{0.90})$$

where $q_{0.90}$is the 90th percentile of TOTEXPY2 in the analytic sample. This yields **782** positives (≈10%) and **7,030** negatives.

- **Acute utilization targets**
    - **ANY_ED_Y2** (any emergency department visit in Year-2) = 1 if **ERTOTY2 > 0**, else 0 (**1,115** positives; ≈14.3%)
    - **ANY_IP_Y2** (any inpatient stay in Year-2) = 1 if **IPDISY2 > 0**, else 0 (**567** positives; ≈7.3%)

These class balances match the Phase I EDA results and confirm that ED and inpatient outcomes are moderately imbalanced classification tasks.

5.3 Feature engineering and feature set definition

Feature engineering is designed to use **Year-1 predictors only** (baseline information) to avoid leakage when predicting Year-2 outcomes. The feature pipeline generates both (i) interpretable categorical labels for low-cardinality variables and (ii) numeric engineered features (log transforms, binary indicators, multimorbidity).

5.3.1 Candidate features

After excluding identifiers, weights, and targets, the engineered table contains **127** candidate predictors, based on the 103 core variables selected in section 3.4. Data types are used to split features into:

- **Categorical columns (7):**
  AGE_GROUP, RACE_ETH, REGIONY1_CAT, EDU_GROUP, POVCATY1_CAT, FAMSIZE_Y1_GRP, INS_TYPE_Y1 [1]
- **Numeric columns (120):**
  These include both raw MEPS columns and engineered numeric features.

5.3.2 Parsimonious modelling features (engineered over raw)

To keep models stable and interpretable, we do not feed all raw numeric codes into the baseline models. Instead, we build a parsimonious numeric set that prioritizes engineered, Year-1 predictors such as:

- demographics/SES: AGE, SEX_BIN, LOG_FAMINCY1, FAMSIZE_Y1
- baseline utilization/cost: LOG_TOTEXPY1, ANY_ED_Y1, ANY_IP_Y1
- health status: RTHLTH1_FAIRPOOR, MNHLTH1_FAIRPOOR
- chronic conditions: condition flags (*_BIN) + multimorbidity score
- employment: WORKED_Y1, unemployment compensation indicators, and an “information available” flag to distinguish NIU/unknown from true zeros

Categorical variables are mapped to labels before one-hot encoding. (e.g., INS_TYPE_Y1, POVCATY1_CAT) so they can be encoded consistently in the modelling pipeline.

5.4 Model-ready encoding and imputation

For modelling, we use a scikit-learn preprocessing pipeline based on a ColumnTransformer:

- **Numeric preprocessing:** median imputation (SimpleImputer(strategy="median"))
- **Categorical preprocessing:** most-frequent imputation (SimpleImputer(strategy="most_frequent")) followed by one-hot encoding (OneHotEncoder(handle_unknown="ignore"))

This produces a numeric design matrix X suitable for linear models and classifiers while ensuring that unseen categories in the validation/test splits do not cause runtime errors. In Phase III, preprocessing steps are fit on the training split and applied to validation/test splits as part of the end-to-end pipeline.

5.5 Output of Phase II

Phase II outputs a single modelling table (df_feat) that contains:

- cleaned Year-1 features (engineered numeric features + labeled categoricals),
- Year-2 targets for regression and classification

This dataset is used as the input for all baseline and comparative models in Phase III.

6. Phase III: Modelling

This phase evaluates predictive models that use **Year-1 information** to forecast **Year-2 outcomes**. All models are trained and evaluated using a consistent train/validation/test protocol with a shared preprocessing pipeline (imputation + one-hot encoding for categoricals). Hyperparameters and decision thresholds (for classification) are selected using the validation set only.

[1] Definitions of variables and acronyms are provided in Appendix D.

6.1 Targets & Evaluation Metrics

Targets

The modelling tasks follow the targets defined in Section 5.2. In brief, we consider:

- **Regression:** Year-2 total expenditures on the log scale (LOG_TOTEXPY2).
- **Classification:** three binary outcomes in Year-2: high-cost status (HIGHCOST_Y2), any ED visit (ANY_ED_Y2), and any inpatient stay (ANY_IP_Y2).

Evaluation metrics

- **Regression (LOG_TOTEXPY2):** MAE on log scale, RMSE on log sale, and R^2 on the validation/test sets.
- **Classification:** ROC AUC and PR AUC (Average Precision). Because the outcomes are imbalanced (especially inpatient stays), PR AUC is reported alongside ROC AUC. In addition, a decision threshold is selected on the validation set to maximize F1; that threshold is then applied unchanged to the test set.

6.2 Baselines

Baseline setup

All baselines use the same pipeline structure:

- **Predictors:** a parsimonious feature set drawn from demographics, SES, insurance, employment, baseline health, chronic conditions, and Year-1 utilization/cost. Two feature configurations are evaluated: a **key predictor set** (AGE, LOG_TOTEXPY1, ANY_ED_Y1, ANY_IP_Y1, INS_TYPE_Y1) and the **full candidate feature set**; both use the same pipeline for a fair comparison.
- **Preprocessing:** numeric imputation (median); categorical imputation (most frequent) + one-hot encoding (handle_unknown="ignore").
- **Splitting:** train/validation/test split with validation used for model selection. Classification splits use stratification to preserve class proportions.

Baseline models

- **Regression baseline:** ElasticNet predicting LOG_TOTEXPY2.
- **Classification baselines:** Logistic regression with class balancing for HIGHCOST_Y2, ANY_ED_Y2, and ANY_IP_Y2. For each classification model, the decision threshold is chosen on the validation set to maximize F1.

Baseline performance

Table 6.2-1. Baseline model performance

Task	Feature set	Baseline model	Test metric(s)	Notes
Expenditure regression	Key predictors	ElasticNet	R^2 = 0.490; RMSE_log = 2.239	log target
Expenditure regression	Full features	ElasticNet	R^2 = 0.523; RMSE_log = 2.165	
High-cost prediction	Key predictors	Logistic (class_weight=balanced)	AUC = 0.863; PR-AUC = 0.466; F1 = 0.530	Strongest classification baseline
High-cost prediction	Full features	Logistic (class_weight=balanced)	AUC = 0.850; PR-AUC = 0.424; F1 = 0.450	
Any ED visit	Key predictors	Logistic (class_weight=balanced)	AUC = 0.698; PR-AUC = 0.314; F1 = 0.319	Moderate discrimination

Any ED visit	Full features	Logistic (class_weight=balanced)	AUC = 0.707; PR-AUC = 0.331; F1 = 0.373	
Any inpatient stay	Key predictors	Logistic (class_weight=balanced)	AUC = 0.752; PR-AUC = 0.203; F1 = 0.269	Most challenging
Any inpatient stay	Full features	Logistic (class_weight=balanced)	AUC = 0.751; PR-AUC = 0.220; F1 = 0.277	

PR-AUC summarizes the precision–recall trade-off (precision = TP/(TP+FP), recall = TP/(TP+FN)) and is more informative than ROC-AUC when the positive class is rare.

A small set of key predictors set (age, baseline cost, baseline utilization, insurance type) already provides strong baseline performance. For regression, expanding to the full candidate feature set yields a modest but consistent improvement (lower RMSE_log and higher R^2), while for classification, expanding to the full candidate feature set yields only modest gains for ED and inpatient prediction and does not improve HIGHCOST discrimination, indicating that the main signal is captured by core utilization and cost variables.

Classification achieves higher AUC/PR-AUC, while regression is evaluated separately by R^2/MAE, which is expected given the heavy-tailed and highly heterogeneous nature of annual costs. We therefore emphasize ranking-based evaluation for imbalanced events (PR-AUC in addition to ROC-AUC). PR-AUC reflects ranking quality under class imbalance. We also apply a top-k framing for capacity-constrained settings (limited care-management slots / limited intervention capacity).

6.3 Class Imbalance Handling

All three classification targets are imbalanced (especially **ANY_IP_Y2**). To address this, we used a combination of **metric choice, sampling strategy, weighting, and thresholding**:

6.3.1 Imbalance-aware evaluation

Accuracy is not informative under imbalance. Therefore, model selection emphasized:

- **PR-AUC** (Average Precision) as the primary selection metric for classification, since it is sensitive to minority-class performance.
- **ROC-AUC** as a secondary metric for ranking quality.
- **F1** at a tuned threshold to summarize a balanced precision–recall operating point.

6.3.2 Stratified splitting

Train/validation/test splits for classification were performed with **stratification**, maintaining the class proportion across splits (stratified split preserving the 60:20:20 ratio.) and reducing instability in validation metrics.

6.3.3 Cost-sensitive training (weights)

Because different algorithms support class imbalance in different ways, we applied cost-sensitive weighting using each model's standard mechanism:

- **Random Forest:** class_weight="balanced_subsample" to reduce majority-class dominance during tree construction.
- **XGBoost:** for rare outcomes, imbalance can be handled with

  $$scale\_pos\_weight = \frac{n_{neg}}{n_{pos}}$$

  to up-weight positive class errors during training.
- **MLP:** balanced sample_weight was used to reduce bias toward predicting the majority class.

6.3.4 Threshold tuning and risk stratification

Even with weighting, the choice of the decision threshold t strongly determines precision vs recall.

We therefore reported:

1. **F1-optimal threshold** (chosen on validation) for a balanced operating point, and
2. **Risk stratification ("Top-k")** views when appropriate, where the top 5/10/20% highest predicted risks are flagged. This approach matches realistic resource constraints (e.g., outreach programs can only target a fixed proportion of the population).

This combination provides a clearer and more actionable view of model utility under class imbalance than a single default threshold of 0.5.

6.4 Model Predictions

After defining the imbalance-aware evaluation framework, we compared Random Forest, XGBoost, and MLP because they represent three common paradigms for tabular prediction: Random Forest as a robust nonlinear baseline, boosted trees (XGBoost) as a strong method for structured data, and a neural network (MLP) to test whether a flexible deep model adds value beyond tree-based approaches.

6.4.1 Experimental setup

All prediction models were trained using **Year-1 features** to predict **Year-2 outcomes**, following the same pipeline to ensure fair comparison:

- **Train/validation/test split** (classification stratified by the target).
- **Preprocessing**: median imputation for numeric features and most-frequent imputation for categorical features; categorical variables were one-hot encoded. For MLP, numeric features were additionally **standardized**.
- **Model selection**: hyperparameters were selected on the **validation set**.
- **Evaluation**: final performance was reported on the **held-out test set** (single evaluation after model selection).

For classification tasks, models output a predicted probability $p_i = P(Y = 1 \mid X_i)$. A probability threshold t is then used to convert probabilities into class labels. In the main results below, the classification threshold was selected on the validation set to **maximize F1** and then applied to the test set.

6.4.2 Regression: Year-2 log total expenditure (LOG_TOTEXPY2)

We compared Random Forest (RF), XGBoost (XGB) and a Multilayer Perceptron (MLP) to predict **LOG_TOTEXPY2**. Overall, all three models achieved **very similar** test performance (R^2 around 0.53), suggesting diminishing returns from increasing model complexity once expenditures are log-transformed.

- **XGB with early stopping + stronger regularization** produced the best (or near-best) **RMSE/R^2**, while
- **RF** tended to have slightly better **MAE** (more robust average error),
- **MLP** was comparable, with slightly worse MAE than tree-based models.

Table 6.4-1. LOG_TOTEXPY2

Model	RMSE_log	MAE_log	R^2
Random Forest (tuned)	2.153	1.533	0.529
XGBoost (early stopping)	**2.150**	**1.542**	**0.530**
MLP	2.149	1.547	0.530

In practice, these differences are small. Therefore, the regression results support the conclusion that multiple non-linear models can reach a similar level of predictive accuracy for log-expenditures under the current feature set. For the thesis, **XGB is the primary model for LOG_TOTEXPY2.**

6.4.3 Classification: HIGHCOST_Y2, ANY_ED_Y2, ANY_IP_Y2

Table 6.4-2. HIGHCOST_Y2 (prevalence ≈ 10.0%)

Model	AUC	PR-AUC	F1@t*	t*	Precision@t*	Recall@t*	TN	FP	FN	TP
Random Forest	**0.868**	**0.459**	**0.536**	**0.55**	**0.487**	**0.596**	**1309**	**98**	**63**	**93**
XGBoost	0.856	0.443	0.525	0.65	0.444	0.641	1282	125	56	100
MLP	0.853	0.420	0.459	0.75	0.397	0.545	1278	129	71	85

Random Forest provides the best overall performance on this target, with the highest PR-AUC (0.459) and F1 (0.536). XGBoost increases recall (0.641 vs 0.596) but at the cost of more false positives and lower precision, resulting in slightly lower F1. MLP is clearly behind tree-based models. For the thesis, **RF is the primary model for HIGHCOST_Y2**.

Table 6.4-3. ANY_ED_Y2 (prevalence ≈ 14.3%)

Model	AUC	PR-AUC	F1@t*	t*	Precision@t*	Recall@t*	TN	FP	FN	TP
XGBoost	**0.728**	**0.350**	**0.366**	**0.20**	**0.313**	**0.439**	**1125**	**215**	**125**	**98**
Random Forest (tuned)	0.707	0.320	0.362	0.40	0.312	0.430	1128	212	127	96
MLP	0.710	0.328	0.344	0.60	0.290	0.422	1110	230	129	94

Although thresholded classification outcomes are similar, XGB's higher PR-AUC suggests it provides a better **risk ordering**, which is useful for **risk stratification**.

Risk stratification (Top-k policy) using RF probabilities
To reflect resource-constrained settings, we also evaluated "Top-k" selection for ED risk. Under RF:

- Top 10% highest risk: precision 0.410, recall 0.287
- Top 20% highest risk: precision 0.313, recall 0.439

Notably, the "Top 20%" policy closely matches the F1-optimal threshold operating point (similar flagged counts and recall), providing an intuitive implementation framework for intervention targeting.

Given higher PR-AUC and AUC, **XGB is selected as the primary model for ANY_ED_Y2**.

Table 6.4-4. ANY_IP_Y2 (prevalence ≈ 7.26%)

Model	AUC	PR-AUC	F1@t*	t*	Precision@t*	Recall@t*	TN	FP	FN	TP
Random Forest (tuned)	**0.755**	**0.225**	**0.285**	**0.40**	**0.234**	**0.363**	**1316**	**134**	**72**	**41**
XGBoost	0.752	0.202	0.261	0.15	0.204	0.363	1290	160	72	41
MLP	0.758	0.202	0.275	0.65	0.215	0.381	1293	157	70	43

Both tree models capture the same number of true admissions (**TP=41**) and have identical recall at their chosen thresholds, but **XGB produces more false positives** (FP=160 vs 134), lowering precision and reducing F1. **MLP** achieves a slightly higher recall (**TP=43**, recall **0.381** vs 0.363) but also flags more individuals (FP=157), resulting in **precision 0.215** and **F1=0.275**, which remains below the tuned RF baseline (**F1=0.285**). Overall, RF offers the best trade-off for this rare outcome, consistent with its higher PR-AUC (**0.225** vs **0.202** for XGB/MLP).

Risk stratification (Top-k policy) using RF probabilities
Because inpatient admission is rare and interventions are typically capacity-limited, Top-k evaluation provides a practical lens. Under RF:

- Top 5% highest risk: precision 0.333, recall 0.230
- Top 10% highest risk: precision 0.237, recall 0.327
- Top 20% highest risk: precision 0.182, recall 0.504

This shows that selecting the top 20% highest-risk individuals captures roughly half of inpatient admissions, although precision decreases as the flagged population grows.

Based on PR-AUC/F1 and a more favorable false-positive burden at comparable recall, **RF is selected as the primary model ANY_IP_Y2**.

7. Phase IV: Results & Error Analysis

Phase IV summarizes predictive performance across targets and provides a brief error analysis to clarify how models should be used.

Results:

Table 7-1. Final model performance by target (test set)

Target	*Prevalence (proportion of positives in the sample)*	*Primary model*	*AUC*	*PR-AUC*	*F1@t**	*t**	*Notes*
HIGHCOST_Y2	~10.0%	Random Forest	0.868	0.459	0.536	0.55	Strongest discrimination
ANY_ED_Y2	~14.3%	XGBoost	0.728	0.350	0.366	0.20	Moderate, best ranking quality
ANY_IP_Y2	~7.3%	Tuned Random Forest	0.755	0.225	0.285	0.40	Hardest (rare outcome)
LOG_TOTEXPY2	—	XGBoost (early stopping)	—	—	—	—	RMSE_log=2.15, R^2=0.53

Table 7-1 summarizes test performance across targets. HIGHCOST_Y2 achieves the strongest discrimination, ED is moderately predictable, and inpatient admission remains the most challenging due to lower prevalence. For regression, the early-stopped XGBoost model achieves consistent but imperfect accuracy ($R^2 \approx 0.53$), indicating Year-1 features explain a meaningful yet incomplete share of Year-2 expenditure variation.

Error Analysis:

We focus error analysis on **ANY_IP_Y2 (Year-2 inpatient admission)** because it is the **hardest target** (low prevalence and admissions are often driven by short-term shocks not captured in Year-1 survey variables). Using the final inpatient model at the operating point (validation-selected threshold t=0.40), errors show a clear pattern: predicted risk is high for **TP** and **FP** (mean proba ≈ 0.565 and 0.514), but much lower for **FN** (mean ≈ 0.210), and **no FN exceeds the threshold**

(FN max ≈ 0.392) — threshold = 0.40; all FN have predicted probability < 0.40. This suggests missed admissions are not borderline threshold cases; they appear low-risk given available Year-1 predictors.

Comparing **false positives vs false negatives** indicates that false positives tend to have a **higher baseline risk burden**: they are older (AGE ≈ 68.3 vs 49.5) and have higher prior spending (LOG_TOTEXPY1 ≈ 9.75 vs 7.60), as well as higher rates of prior ED use (ANY_ED_Y1 ≈ 0.403 vs 0.194) and hypertension (HIBPDXY1_BIN ≈ 0.754 vs 0.417). Overall, the inpatient model is best interpreted as a **risk stratification tool** rather than a deterministic predictor of admissions.

Table 7-2. ANY_IP_Y2 error summary (t = 0.40)

Item	Value
Confusion matrix	TN=1316, FP=134, FN=72, TP=41
Mean predicted probability	TP=0.565, FP=0.514, FN=0.210, TN=0.141
FN highest probability	0.392 (< 0.40)
FP vs FN (AGE)	68.3 vs 49.5
FP vs FN (LOG_TOTEXPY1)	9.75 vs 7.60
FP vs FN (ANY_ED_Y1)	0.403 vs 0.194
FP vs FN (HIBPDXY1_BIN)	0.754 vs 0.417

8. Phase V: Explainability & Fairness

8.1 Drivers of Predictions: Feature Contribution Analysis

To explain what drives model predictions, we combine **permutation importance** (global contribution ranking) with a limited set of **directional subgroup summaries** (observed rates/means and mean model-assigned risk/mean predictions). For classification targets, feature contribution is quantified as the **mean decrease in PR-AUC (average precision)** when a feature is randomly permuted (shuffled). For the expenditure regression, contribution is quantified as the **mean increase in RMSE_log** after permutation. Because detailed subgroup breakdowns can be lengthy, the main text reports only the most salient contrasts, while full tables are provided in **Appendix B**.

8.1.1 Global feature contribution (permutation importance)

Table 8.1-1 summarizes the strongest drivers for each target under the selected primary model. Across all targets, **baseline spending and prior utilization** repeatedly appear among the top contributors, consistent with persistence in health care need and utilization patterns. For HIGHCOST_Y2 and LOG_TOTEXPY2, prior-year spending dominates; for ED and inpatient outcomes, prior utilization and broad baseline risk markers remain central.

Table 8.1-1. Top drivers by target (permutation importance).
Classification: mean decrease in PR-AUC; Regression: mean increase in RMSE_log.

Target (Primary model)	Importance metric	Top drivers (value)
HIGHCOST_Y2 (RF)	ΔPR-AUC	LOG_TOTEXPY1 (0.116), AGE (0.045), ANY_ED_Y1 (0.031), HIBPDXY1_BIN (0.030), FAMSIZE_Y1_GRP (0.023)
ANY_ED_Y2 (XGB)	ΔPR-AUC	LOG_TOTEXPY1 (0.084), ANY_ED_Y1 (0.053), LOG_FAMINCY1 (0.039), AGE (0.037), LOG_UNEMP_COMP_Y1 (0.018)
ANY_IP_Y2 (RF)	ΔPR-AUC	LOG_TOTEXPY1 (0.046), ANY_ED_Y1 (0.028), HIBPDXY1_BIN (0.015), AGE (0.014), SEX_BIN (0.007)

LOG_TOTEXPY2 (XGB-ES)	ΔRMSE_log	LOG_TOTEXPY1 (0.996), AGE (0.056), RACE_ETH (0.050), INS_TYPE_Y1 (0.033), FAMSIZE_Y1 (0.021)

Full variable names and definitions are provided in Appendix D (Variable Glossary).

8.1.2 Directional interpretation (selected contrasts)

Permutation importance identifies which features matter most but does not directly indicate direction. To provide directional context, we additionally examined **subgroup contrasts** for a small set of clinically meaningful variables. Specifically, for each target we report (i) the **observed subgroup outcome rate/mean** and (ii) the model's **mean predicted risk** (classification) or **mean predicted outcome** (regression) within the same subgroup. **Observed subgroup rates are computed on the full analytic sample, whereas model-assigned risk is summarized on the held-out test split only.** Table 8.1-2 reports the most informative contrasts for each target. Full feature rankings and complete subgroup tables are reported in **Appendix B (Tables B1–B4)**.

Table 8.1-2. Key directional patterns (selected subgroup contrasts).
Observed = observed outcome prevalence rate/mean; Model = mean predicted risk/mean prediction.

Target	Key contrast	Observed (high vs low risk group)	Model (high vs low risk group)
HIGHCOST_Y2	LOG_TOTEXPY1 Q4 vs Q1 (top vs bottom quartile of LOG_TOTEXPY1, Q4=top 25%)	29.4% vs 1.6%	53.9% vs 7.4%
	ANY_ED_Y1=1 vs 0	22.0% vs 8.1%	44.9% vs 21.9%
	AGE 65+ vs 0–17	20.9% vs 2.0%	44.2% vs 7%
ANY_ED_Y2	ANY_ED_Y1=1 vs 0	33.6% vs 11.1%	32.4% vs 11.1%
	LOG_TOTEXPY1 Q4 vs Q1	25.6% vs 6.4%	24.7% vs 5.8%
	AGE 65+ vs 18–29	21.6% vs 9.8%	21.9% vs 10.5%
ANY_IP_Y2	LOG_TOTEXPY1 Q4 vs Q1	16.0% vs 2.3%	36.2% vs 8.1%
	AGE 65+ vs 0–17	15.7% vs 1.8%	36.7% vs 7.9%
	HIBPDXY1=1 vs 0	14.3% vs 4.2%	32.0% vs 13.6%
LOG_TOTEXPY2	LOG_TOTEXPY1 Q4 vs Q1	mean 9.02 vs 3.57	mean 9.03 vs 3.60
	AGE 65+ vs 18–29	mean 8.48 vs 5.00	mean 8.47 vs 4.89

8.1.3 Target-specific interpretation

HIGHCOST_Y2. High-cost risk is most strongly driven by **prior-year spending**, with additional stratification from age, baseline utilization (ED/IP), and chronic burden/health status. Subgroup patterns show clear concentration among **older adults** and **high baseline spenders,** supporting the interpretation of persistence in high-cost status.

ANY_ED_Y2. ED use exhibits a strong **utilization persistence effect**: individuals with **higher baseline spending** have substantially higher Year-2 ED prevalence and predicted risk. Having an ED visit in Year-1 (ANY_ED_Y1=1) and older age further separate higher-risk from lower-risk groups.

ANY_IP_Y2. Inpatient admission is primarily driven by **baseline risk proxies (prior spending and age)** and **chronic burden** (e.g., hypertension). The prominence of ANY_ED_Y1 suggests prior ED use is a marker of health instability that also increases subsequent hospitalization risk. As with ED, the inpatient model is best interpreted as a **risk stratification tool** for a rare outcome.

LOG_TOTEXPY2 regression. Expenditure predictions are dominated by **LOG_TOTEXPY1**, confirming strong temporal persistence in spending; age provides smaller but consistent additional stratification. The model reproduces clear gradients across prior-year spending quartiles and age groups in both observed and predicted mean log expenditures.

Finally, these results should be interpreted as **associations learned from Year-1 survey variables rather than causal effects**. Differences across subgroups may be driven by underlying variation in health needs, access to care, and typical patterns of healthcare use.

8.2 Fairness audit executive summary

We conducted a lightweight fairness audit across **sex** and **age bands** using group-wise **PR-AUC** and **recall under top-k targeting** (top 10% / 20% highest-risk). Results suggest **limited sex-based differences** for HIGHCOST_Y2 and ANY_IP_Y2 under capacity-limited targeting (e.g., HIGHCOST recall@top10% ≈ 0.54 for both males and females; ANY_IP recall@top10% ≈ 0.34 vs 0.32), while **ANY_ED_Y2 shows larger sex differences** in ranking and capture (higher PR-AUC and top-k recall among females). In contrast, **age-related heterogeneity is substantial across outcomes**, with consistently stronger capture in older adults and near-zero recall for very low-prevalence pediatric groups.

Table 8.2-1a Fairness audit by sex (PR-AUC and recall@top10%)

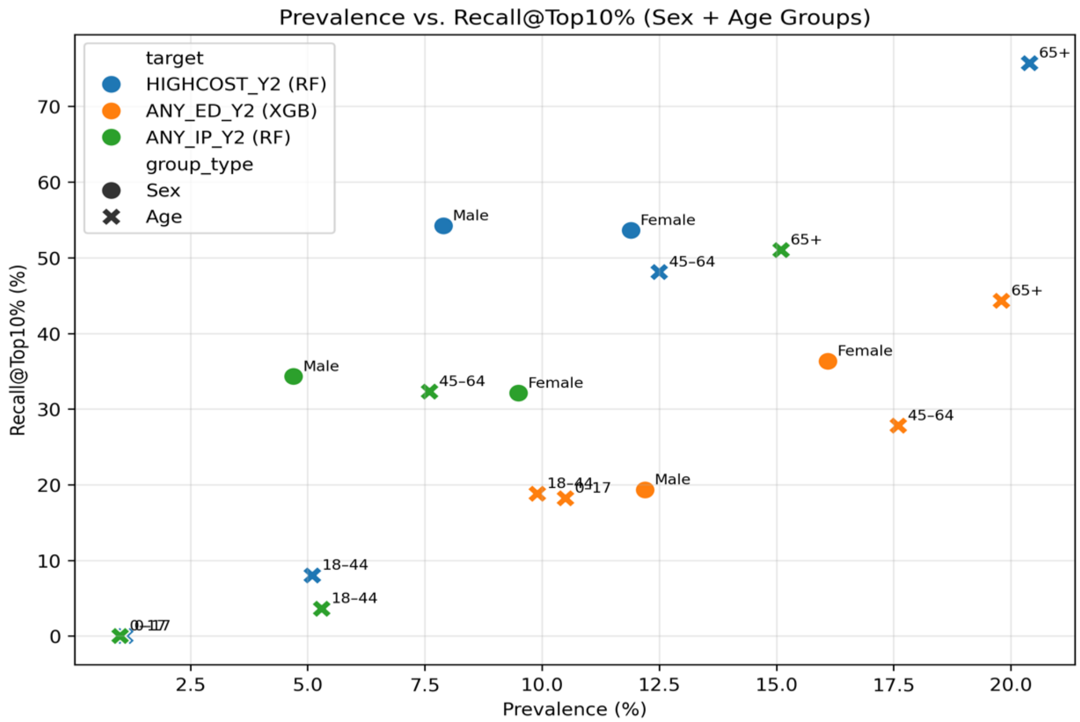
Target	Sex	n	Prevalence	PR-AUC	Recall@Top10%
HIGHCOST_Y2 (RF)	Male	751	7.9%	0.497	54.2%
HIGHCOST_Y2 (RF)	Female	812	11.9%	0.444	53.6%
ANY_ED_Y2 (XGB)	Male	722	12.2%	0.330	19.3%
ANY_ED_Y2 (XGB)	Female	841	16.1%	0.365	36.3%
ANY_IP_Y2 (RF)	Male	740	4.7%	0.190	34.3%
ANY_IP_Y2 (RF)	Female	823	9.5%	0.250	32.1%

Table 8.2-1b Fairness audit by age group (PR-AUC and recall@top10%)

Target	Age group	n	Prevalence	PR-AUC	Recall@Top10%
HIGHCOST_Y2 (RF)	0–17	281	1.1%	0.394	0.0%
HIGHCOST_Y2 (RF)	18–44	486	5.1%	0.211	8.0%
HIGHCOST_Y2 (RF)	45–64	433	12.5%	0.485	48.1%
HIGHCOST_Y2 (RF)	65+	363	20.4%	0.523	75.7%
ANY_ED_Y2 (XGB)	0–17	315	10.5%	0.345	18.2%
ANY_ED_Y2 (XGB)	18–44	485	9.9%	0.316	18.8%
ANY_ED_Y2 (XGB)	45–64	409	17.6%	0.394	27.8%
ANY_ED_Y2 (XGB)	65+	354	19.8%	0.365	44.3%

ANY_IP_Y2 (RF)	0–17	287	1.0%	0.018	0.0%
ANY_IP_Y2 (RF)	18–44	528	5.3%	0.091	3.6%
ANY_IP_Y2 (RF)	45–64	410	7.6%	0.277	32.3%
ANY_IP_Y2 (RF)	65+	338	15.1%	0.286	51.0%

Figure 8.2-1 Prevalence vs. Recall@Top10% (Sex + Age Groups)



9.Phase VI: Productization Concept: Health Outcome Predictor (MVP)

9.1 What the MVP does

To demonstrate how the thesis models could be used in a capacity-constrained setting, we built a lightweight Streamlit application, **Health Outcome Predictor (MVP)**. The MVP performs **inference only** (no retraining) and supports two operational tasks:

- **Classification (risk ranking):** score individuals with a predicted probability for Year-2 binary outcomes (e.g., high-cost, ED use, inpatient use).
- **Regression (cost ranking):** predict Year-2 total expenditure on the original scale in **USD**, using the inverse transform of the thesis target:

$$\widehat{TOTEXPY2} = exp\,(log\,(1+\widehat{TOTEXPY2})) - 1$$

The core usage is **top-k selection**: the user chooses a fraction (e.g., top 10%), and the app returns the ranked list and a selection flag for the top-k individuals.

9.2 Inputs and outputs

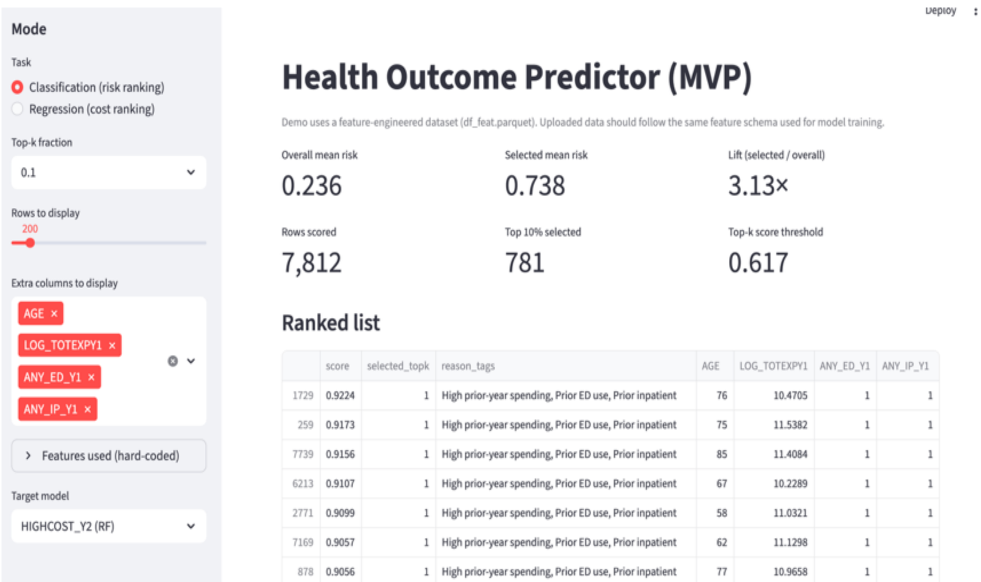
Input. The demo mode uses df_feat.parquet, a feature-engineered dataset derived from Year-1 variables only. Uploaded data is expected to follow the same feature schema (feature-level input).

Output. The app displays:

- number of rows scored and selected (k),
- **overall vs selected** mean score/cost,
- **lift** = (selected mean)/(overall mean),
- a **top-k threshold** (minimum score/cost among the selected group),
- a ranked table with downloadable CSV (including prediction and selected_topk).

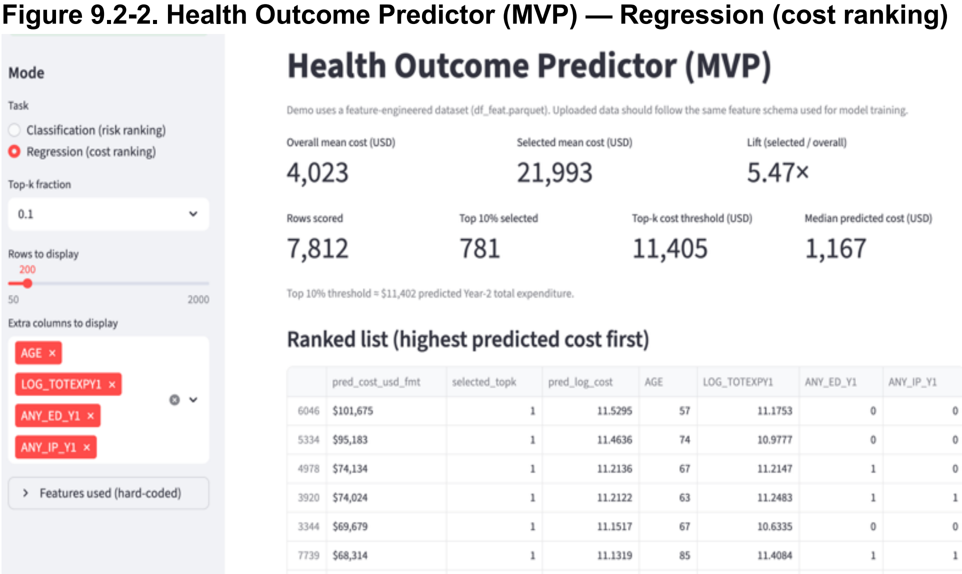
To improve readability of the ranked list, the MVP also shows **rule-based reason tags** (e.g., prior ED use, high prior-year spending). These tags are simple, rule-based summaries of selected baseline variables to improve readability of the ranked list. SHAP-style local attributions could be added in future work.

Figure 9.2-1. Health Outcome Predictor (MVP) — Classification (risk ranking)



The interface scores individuals using the trained classification model, supports capacity-constrained **top-k selection**, and reports overall vs selected mean risk, lift, and the top-k score threshold.

Figure 9.2-2. Health Outcome Predictor (MVP) — Regression (cost ranking)



The interface ranks individuals by predicted Year-2 total expenditure on the USD scale (inverse transform from log1p(TOTEXPY2)), supports top-k selection, and reports overall vs selected mean cost, lift, and the top-k cost threshold. Results are exportable as a CSV.

9.3 Reproducibility

The implementation is provided in the GitHub repository: < https://github.com/WenxiGu/Thesis-project-MEPS > (see README for setup and execution instructions).

10. Phase VII: Conclusions & Business Implications

This thesis evaluated whether Year-1 (baseline) MEPS Panel 27 features can predict Year-2 expenditures and acute events, and whether model performance is consistent across key subgroups. Overall, results support using the models primarily for risk stratification (ranking / top-k selection) rather than deterministic prediction of individual outcomes, reflecting the inherent difficulty of predicting rare events.

RQ1. Which factors are most strongly associated with Year-2 expenditures?

Year-2 expenditure on the log scale (LOG_TOTEXPY2) is driven most strongly by prior-year spending (LOG_TOTEXPY1), indicating strong temporal persistence in costs. Permutation importance for the selected regression model (XGBoost with early stopping) shows LOG_TOTEXPY1 dominating the contribution, with smaller but consistent additional stratification from age and selected socio-demographic variables (e.g., race/ethnicity, insurance type, family size). Subgroup patterns confirm clear gradients: predicted and observed mean log costs increase sharply across LOG_TOTEXPY1 quartiles and are higher among older adults.

Conclusion (RQ1): future spending is predictable mainly because spending is persistent; socio-economic and insurance variables contribute secondary stratification rather than replacing the explanatory role of prior spending.

RQ2. How well can we distinguish high-cost individuals in Year-2?

HIGHCOST_Y2 (top 10% of TOTEXPY2) is the most predictable classification target. The primary Random Forest model achieves AUC=0.868 and PR-AUC=0.459 on the test set, with F1=0.536 at the validation-selected threshold. These results indicate strong ranking quality and practical separability of the high-cost tail using Year-1 predictors.

A survey-weighted robustness check (LONGWT) yields slightly lower AUC/PR-AUC (AUC 0.845, PR-AUC 0.419) but does not change the core conclusion that high-cost stratification remains feasible in a population-representative lens.

Conclusion (RQ2): high-cost risk stratification is operationally viable, especially when implemented as ranking/top-k selection rather than relying on a single probability threshold.

RQ3. Which factors predict Year-2 ED visits and inpatient stays?

ED risk is moderately predictable and shows a strong utilization persistence effect. The selected XGBoost model achieves AUC=0.728 and PR-AUC=0.350 for ANY_ED_Y2, and subgroup contrasts confirm higher observed and predicted ED rates among those with prior ED use, higher baseline spending, and older age.

Inpatient admission is the most challenging target due to lower prevalence (≈7.3%). The tuned Random Forest achieves AUC=0.755 and PR-AUC=0.225 for ANY_IP_Y2. Error analysis indicates false negatives are not borderline threshold cases: FN predicted probabilities are much lower than TP/FP, and no FN exceeds the chosen threshold (FN max ≈ 0.392 < 0.40), consistent with admissions being driven by short-term shocks not captured in baseline survey variables.

Conclusion (RQ3): ED and inpatient outcomes are partly predictable from baseline utilization/spending and health burden, but inpatient prediction should be framed as prioritization rather than event forecasting.

RQ4. Is performance consistent across socio-demographic and/or insurance subgroups?

A lightweight fairness audit suggests limited sex-based differences for HIGHCOST and inpatient targeting under top-k policies, while ED shows more noticeable sex differences (higher PR-AUC and recall@top10% among females). Age-related heterogeneity is substantial: capture is consistently stronger in older groups and near-zero in very low-prevalence pediatric group under small top-k policies, reflecting large prevalence gradients.

Conclusion (RQ4): subgroup performance varies, particularly by **age**. This thesis evaluates fairness only across **age** and **sex**, reinforcing the need to report subgroup metrics and to use risk scores as decision aids with oversight. For subgroups with poorer results (e.g., 0–17), additional targeted data of positive cases in these groups might be needed to train higher recall models. Audits across additional attributes (e.g., race/ethnicity, income, insurance) are reserved for future work.

Business implications

1. **Capacity-based deployment (how to operationalize ranking):**

   In practice, these models are most actionable as ranking tools that support capacity-constrained outreach. A realistic deployment is to generate a monthly prioritized list (top-k) sized to the available care-management capacity (e.g., nurse navigators can contact N members per month). This approach mirrors real-world workflow better than relying on a single fixed probability threshold for rare outcomes.

2. **Implementation detail: fix the outreach budget first → rank members → assign to teams → document reasons and outcomes → review performance quarterly.**

   Target-specific actionability:

- HIGHCOST_Y2 (proactive budgeting & care management): Use the ranking to prioritize members for proactive case management (e.g., medication review, chronic care follow-up, care-plan adherence support).

  KPIs: top-k cost capture; utilization trends, etc.

- ANY_ED_Y2 (navigation and access-focused outreach): Use the ranking to prioritize members for navigation (e.g., scheduling primary care, after-hours alternatives, transportation support where applicable).

  KPIs: Primary Care Provider (PCP) follow-up completion; repeat ED visits among contacted members, etc.

- ANY_IP_Y2: use as a watch-list for clinical review/escalation, not as a claim of preventable admissions.

  KPIs: inpatient capture in top-k; timeliness of follow-up, etc.

3. **Governance and compliance**: Any practical use must incorporate privacy safeguards and comply with MEPS public-use restrictions (no re-identification and no linkage to individually identifiable external datasets).

Limitations and next steps

Model performance is meaningful but not perfect (test $R^2 \approx 0.53$ for costs, and weaker discrimination for ED/IP), which reflects both data limits and the inherent unpredictability of short-term shocks. The analysis uses only MEPS Panel 27 (2022–2023), so results may not fully generalize to other years or settings. MEPS variables are survey-based and self-reported, which introduces measurement error, and the models do not use richer clinical or external data that could improve accuracy. As a result, the models are best suited for **ranking and top-k selection**, not deterministic individual prediction. In terms of model explainability (feature contribution analysis), we used permutation importance to identify global drivers. Permutation importance can be conservative under correlated features and provides global—rather than individual-level—explanations; SHAP-based local explanations are left for future work. Additionally, because permutation importance does not provide direction, we complemented it with a directional analysis that contrasts observed and predicted outcomes across meaningful high- vs low-risk groups (e.g., spending quartiles, prior ED use, age bands). This provides a logical validity check and aligns with the operational use case of risk ranking. Partial dependence plots (PDPs) are left for future work.

Next steps are to check model performance across more groups (e.g., race/ethnicity, income, insurance), verify that predicted probabilities match real-world rates (calibration), monitor whether performance changes over time (drift), and set clear rules for who can access the system and how usage is logged. For very low-prevalence groups (e.g., pediatric inpatient), performance is unstable; improving recall there likely requires more positive cases or targeted data collection.

Appendix A: EDA Supplementary Figures (Phase I)

This appendix provides detailed descriptive figures referenced in Phase I. Figures summarize the analytic sample (YEARIND=1, ALL5RDS=1; n=7,812), baseline predictors, missingness patterns, and outcome distributions. These plots support the modelling choices in Phase II–III (log-transform for costs; binary modelling and ranking metrics for acute events).

Figure A.1. Age distribution (Year-1).
Interpretation: The sample spans the full life course. *Modelling implication:* age is expected to be a strong predictor and is included in all models.

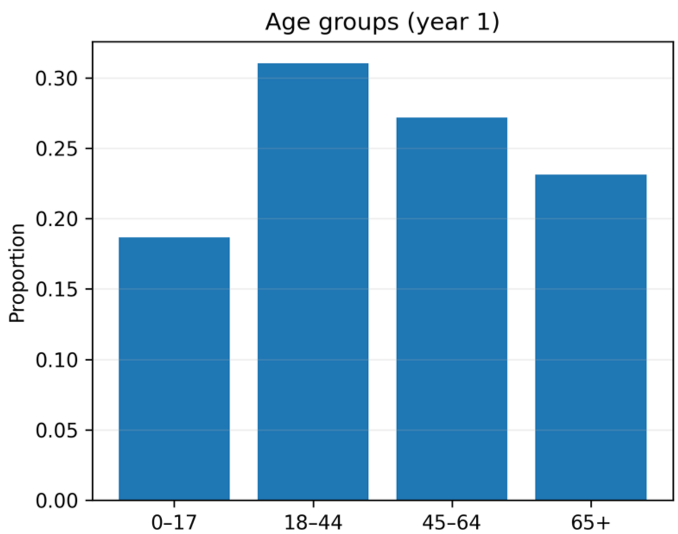
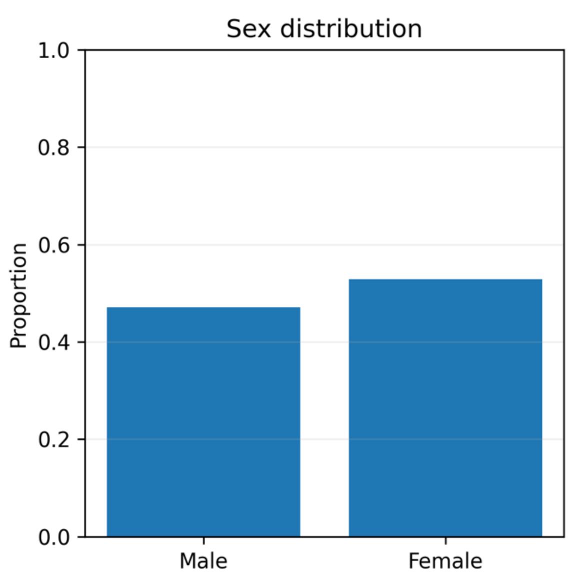
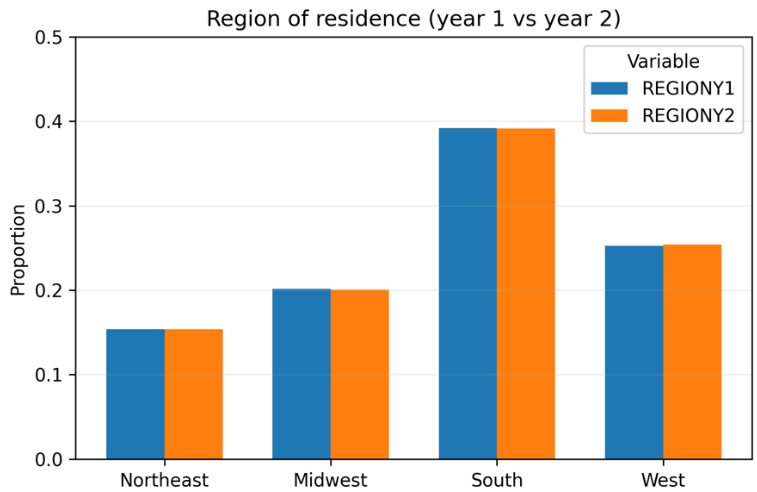


Figure A.2. Sex, race/ethnicity, and region.
Interpretation: The sample is broadly balanced by sex and region, with expected race/ethnicity composition. *Modelling implication:* these variables support subgroup analysis and fairness reporting.





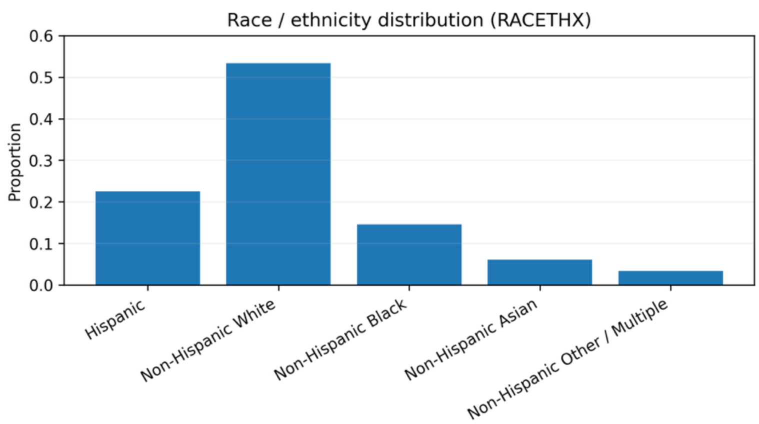
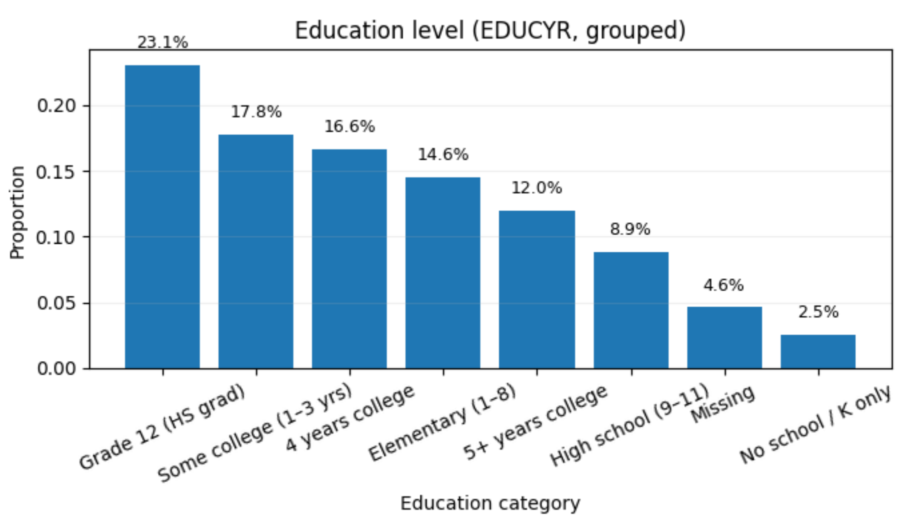
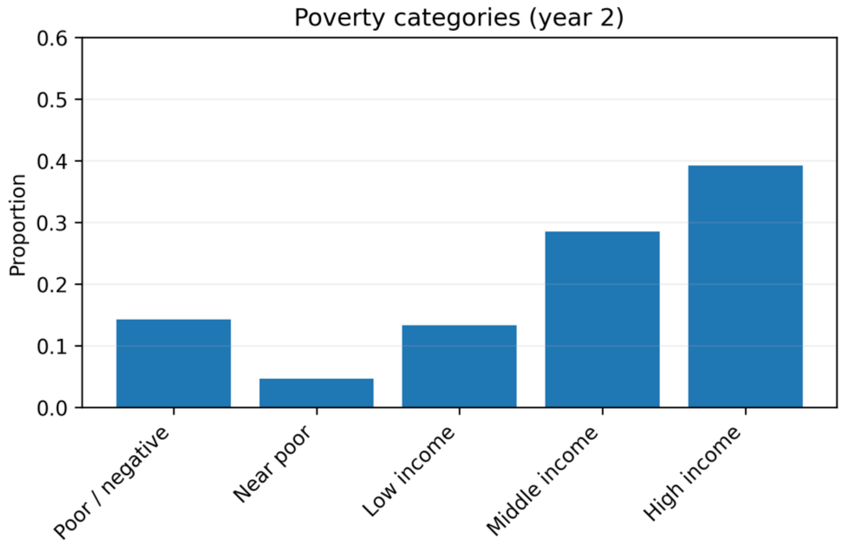


Figure A.3. Education, income and poverty categories.
Interpretation: Income is strongly right-skewed, and education level groups and poverty categories provide a stable SES stratification. *Modelling implication:* SES features are retained as potential drivers of utilization and cost.





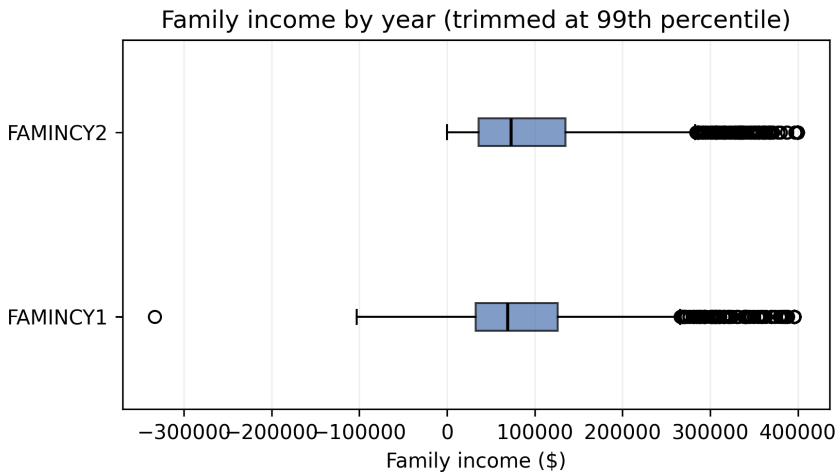
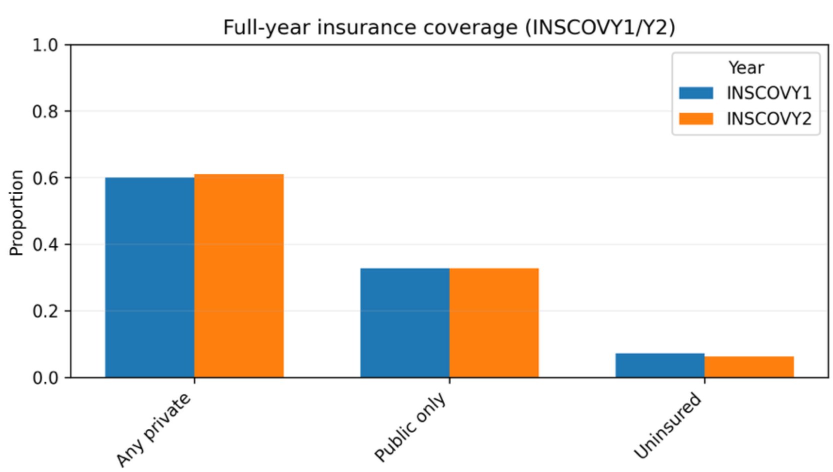


Figure A.4. Insurance coverage and insurance type.
Interpretation: Coverage is relatively stable across years with meaningful variation across private/public/uninsured groups.
Modelling implication: insurance is included as a key policy-relevant predictor.



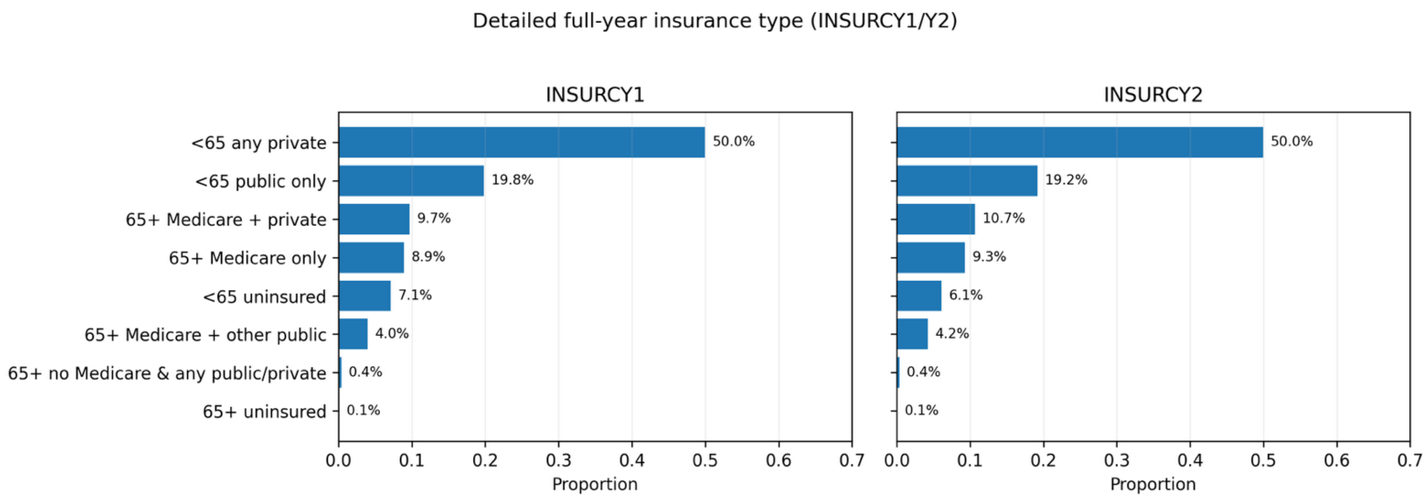
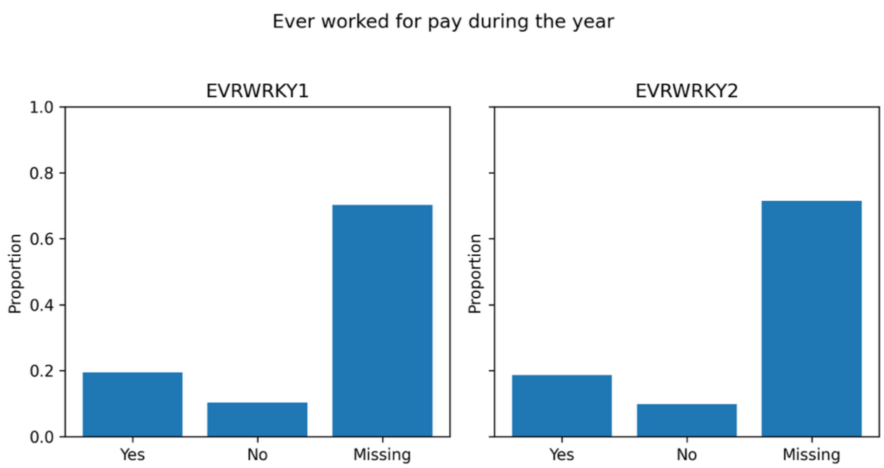
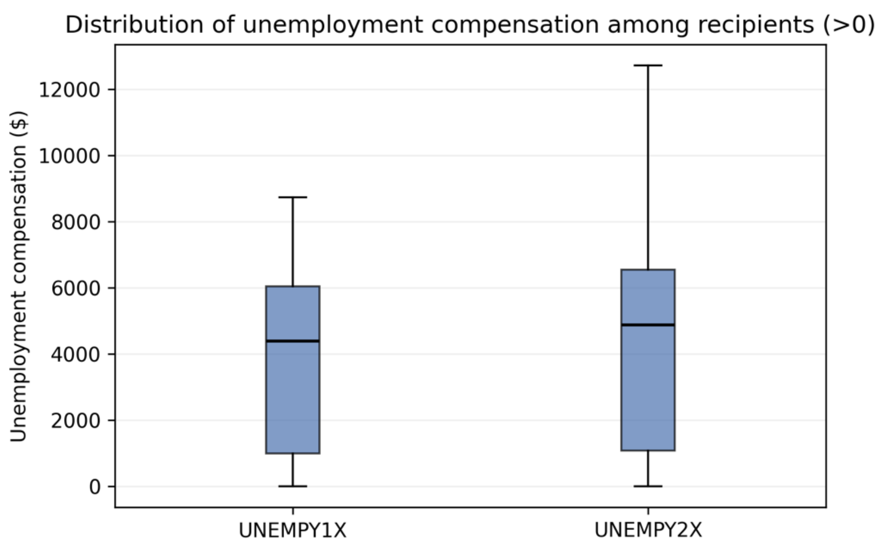


Figure A.5. Employment variables and unemployment compensation.
Interpretation: Many employment fields are inapplicable (children/older adults) and show structural missingness; compensation is highly zero-inflated. *Modelling implication:* employment variables require careful missingness handling and/or engineered indicators





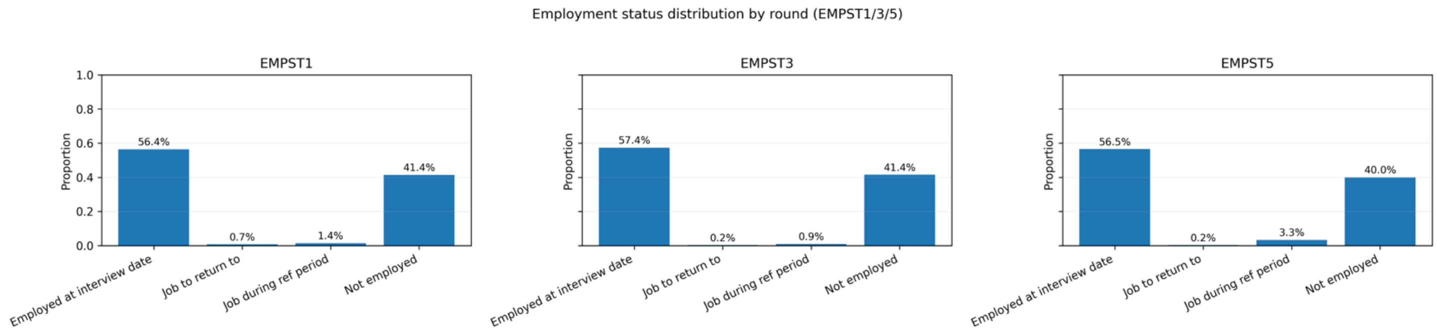


Figure A.6. Self-rated health and mental health.
Interpretation: Most respondents report good-to-excellent health, with a minority reporting fair/poor.

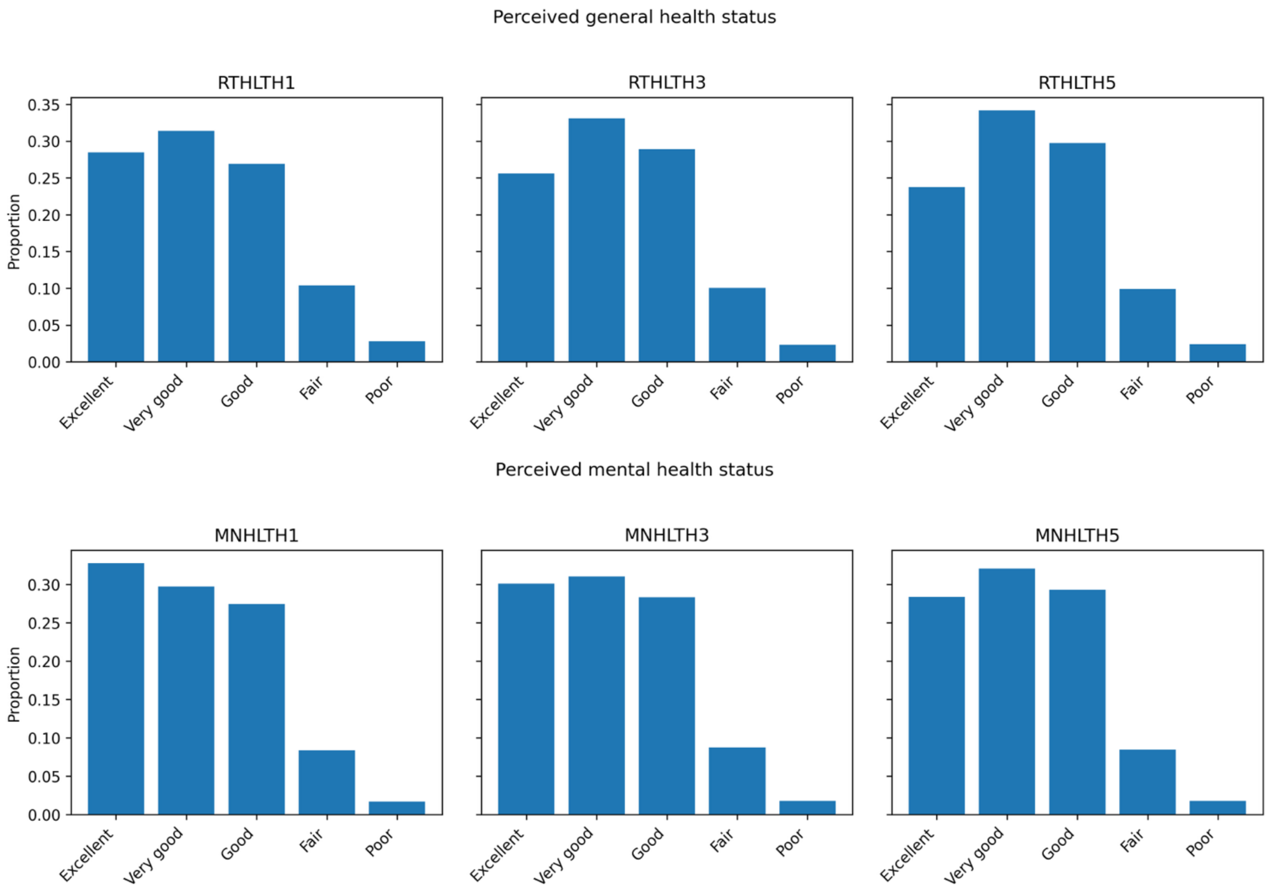


Figure A.7. Chronic condition prevalence and multimorbidity index.
Interpretation: Chronic conditions are common and multimorbidity exists in a non-trivial subgroup.

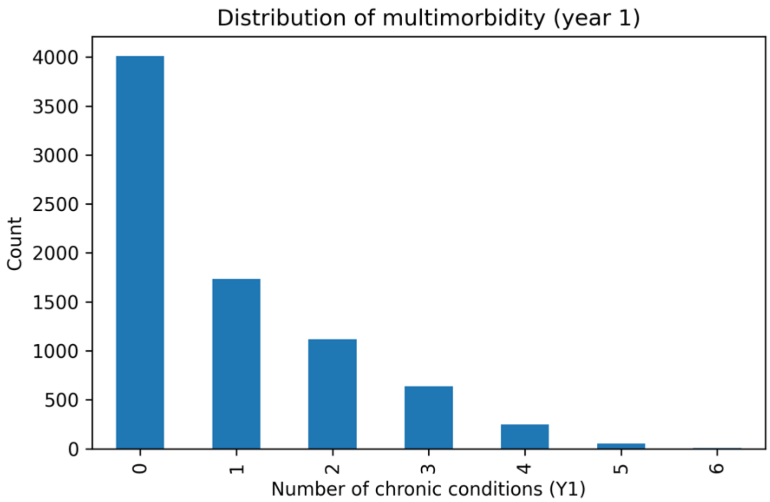
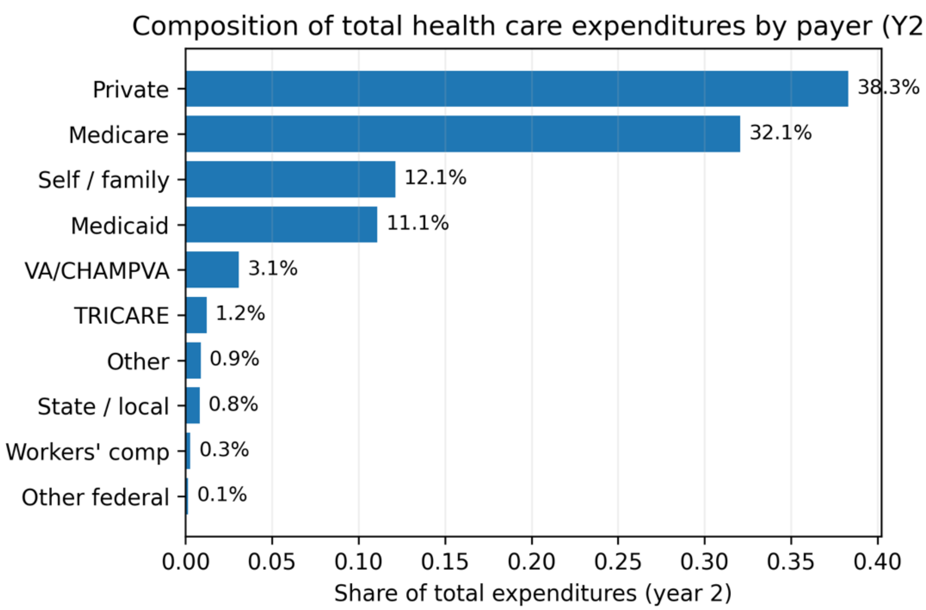
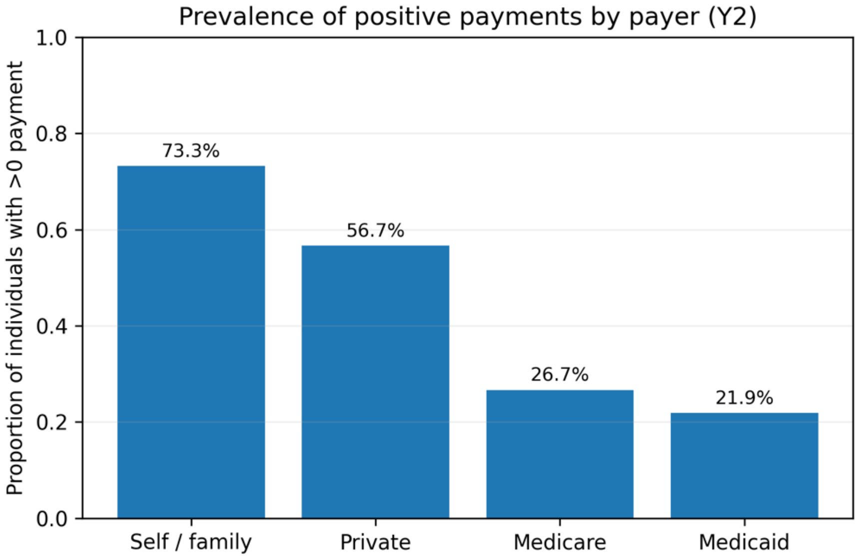
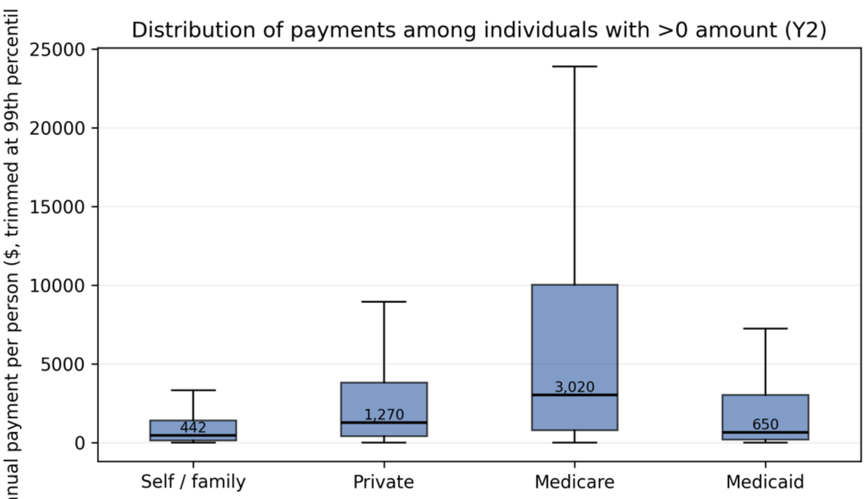


Figure A.8. Payer-specific spending breakdown (Year-2).
Interpretation: Total expenditures are financed primarily through private insurance and Medicare, with smaller contributions from out-of-pocket and Medicaid. From a "who pays more" perspective, Medicare has the highest per-person amounts among those with any Medicare payments, followed by private insurance. Medicaid and out-of-pocket spending tend to have lower per-person amounts, but they have broad coverage across individuals.




Prevalence of positive payments by payer (Y2)
Proportion of individuals with >0 payment
1.0
0.8
0.6
0.4
0.2
0.0
73.3%
56.7%
26.7%
21.9%
Self / family
Private
Medicare
Medicaid


Distribution of payments among individuals with >0 amount (Y2)
nnual payment per person ($, trimmed at 99th percentil
25000
20000
15000
10000
5000
0
442
1,270
3,020
650
Self / family
Private
Medicare
Medicaid

Appendix B: Detailed Feature Contribution and Subgroup Tables

Table B1. HIGHCOST_Y2 subgroup tables (observed rate %, mean predicted risk %)

B1.1 LOG_TOTEXPY1 quartiles

Group	Observed %	Predicted %
Q4(high)	29.4	53.9
Q3	5.9	23.2
Q2	3.1	13.2
Q1(low)	1.6	7.4

B1.2 AGE_GROUP

Group	Observed %	Predicted %
65+	20.9	44.2
45–64	12.2	29.8
30–44	5.9	16.3
18–29	3.0	11.4
0–17	2.0	7.0

B1.3 ANY_ED_Y1

Group	Observed %	Predicted %
1	22.0	44.9
0	8.1	21.9

B1.4 ANY_IP_Y1

Group	Observed %	Predicted %
1	31.5	61.5
0	8.7	22.7

B1.5 RTHLTH1_FAIRPOOR

Group	Observed %	Predicted %
1	24.1	44.7
0	7.9	21.6

B1.6 HIBPDXY1_BIN / CHOLDXY1_BIN

Variable	Group	Observed %	Predicted %
HIBPDXY1_BIN	1	19.8	42.3
HIBPDXY1_BIN	0	5.8	16.4
CHOLDXY1_BIN	1	19.0	40.3
CHOLDXY1_BIN	0	6.4	18.4

B1.7 INS_TYPE_Y1

Group	Observed %	Predicted %
65+ Medicare + other public	28.4	52.4
65+ other coverage	24.1	45.8
65+ Medicare + private	20.3	44.8
65+ Medicare only	18.5	39.9
<65 public only	7.5	18.4
<65 any private	7.1	20.0
<65 uninsured	2.2	7.5
65+ uninsured	0.0	13.6

B1.8 POVCATY1_CAT

Group	Observed %	Predicted %
Near poor	14.0	25.9
Poor/negative	11.6	25.3
Low income	9.4	24.0
Middle income	9.0	22.5
High income	9.8	26.9

B1.9 REGIONY1_CAT / RACE_ETH

Variable	Group	Observed %	Predicted %
REGIONY1_CAT	Northeast	13.5	29.8
REGIONY1_CAT	Midwest	9.5	27.3
REGIONY1_CAT	West	9.0	23.4
REGIONY1_CAT	South	9.5	23.0
RACE_ETH	NH White	12.7	31.0

RACE_ETH	NH Black	9.3	21.3
RACE_ETH	NH Other/multiple	6.7	22.3
RACE_ETH	NH Asian	6.3	16.8
RACE_ETH	Hispanic	5.6	15.4

Table B2. ANY_ED_Y2 subgroup tables (observed % / predicted %)

B2.1 ANY_ED_Y1

Group	Observed %	Predicted %
1	33.6	32.4
0	11.1	11.1

B2.2 AGE_GROUP

Group	Observed %	Predicted %
65+	21.6	21.9
45–64	15.1	14.8
30–44	10.1	9.6
18–29	9.8	10.5
0–17	11.0	9.9

B2.3 LOG_TOTEXPY1 quartiles

Group	Observed %	Predicted %
Q4(high)	25.6	24.7
Q3	14.0	15.0
Q2	11.1	10.5
Q1(low)	6.4	5.8

B2.4 LOG_FAMINCY1 quartiles

Group	Observed %	Predicted %
Q1(low)	20.7	20.1
Q2	14.1	13.5
Q3	11.9	12.1
Q4(high)	10.3	10.7

B2.5 POVCATY1_CAT

Group	Observed %	Predicted %
Near poor	20.2	17.8
Poor/negative	19.1	17.8
Low income	16.0	16.6
Middle income	13.5	13.3
High income	11.5	12.0

B2.6 ANY_UNEMP_COMP_Y1

Group	Observed %	Predicted %
1	24.5	33.0
0	14.1	13.9

B2.7 REGIONY1_CAT

Group	Observed %	Predicted %
Midwest	15.3	15.4
South	15.0	15.1
Northeast	13.5	13.2
West	12.7	12.4

Table B3. ANY_IP_Y2 subgroup tables (observed % / predicted %)

B3.1 LOG_TOTEXPY1 quartiles

Group	Observed %	Predicted %
Q4(high)	16.0	36.2
Q3	5.9	18.5
Q2	4.8	12.8
Q1(low)	2.3	8.1

B3.2 AGE_GROUP

Group	Observed %	Predicted %
65+	15.7	36.7

45–64	7.2	19.4
30–44	4.5	12.5
18–29	4.1	10.9
0–17	1.8	7.9

B3.3 HIBPDXY1_BIN

Group	Observed %	Predicted %
1	14.3	32.0
0	4.2	13.6

Table B4. LOG_TOTEXPY2 regression subgroup tables (mean log cost)

B4.1 LOG_TOTEXPY1 quartiles

Group	Mean actual	Mean predicted
Q4(high)	9.025	9.026
Q3	7.652	7.607
Q2	6.432	6.421
Q1(low)	3.567	3.598

B4.2 AGE_GROUP

Group	Mean actual	Mean predicted
65+	8.477	8.474
45–64	7.185	7.273
30–44	5.792	5.947
0–17	5.640	5.669
18–29	4.999	4.890

Appendix C. Robustness Check: Survey-weighted Performance (LONGWT)

Table C1. Unweighted vs weighted (LONGWT) classification performance

TARGET (PRIMARY MODEL)	T	METRIC	UNWEIGHTED	WEIGHTED (LONGWT)
HIGHCOST_Y2 (RF)	0.55	AUC	0.868	0.845
		PR-AUC	0.459	0.419
		Precision	0.487	0.540
		Recall	0.596	0.325
ANY_ED_Y2 (XGB)	0.20	AUC	0.728	0.720
		PR-AUC	0.350	0.351
		Precision	0.313	0.292
		Recall	0.439	0.433
ANY_IP_Y2 (RF)	0.40	AUC	0.755	0.759
		PR-AUC	0.225	0.190
		Precision	0.234	0.274
		Recall	0.363	0.283

Survey-weighted evaluation was conducted for the classification outcomes to assess population-level robustness under the MEPS longitudinal weights (LONGWT). For the expenditure regression, we report unweighted test performance as the primary evaluation, since the regression model is intended for relative risk/cost ranking rather than population mean estimation, and a full design-based regression evaluation is beyond the scope of this thesis.

We used the same fixed thresholds (T) as in the unweighted evaluation to enable a like-to-like comparison of precision and recall. Because weighting changes the effective population composition, precision–recall trade-offs can shift at a fixed threshold; recalibrating T under weights is possible but was not the goal of this robustness check. As shown in **Table C1**, survey-weighted evaluation (LONGWT) yields slightly lower AUC/PR-AUC across targets compared with unweighted test metrics, indicating that population weighting changes the effective operating point without altering the main conclusions.

Appendix D. Acronyms and Variable Glossary

A. Acronyms

ACRONYM	MEANING
MEPS-HC	Medical Expenditure Panel Survey – Household Component
AHRQ	Agency for Healthcare Research and Quality
HC-252	MEPS Panel 27 longitudinal file (2022–2023)
HC-243	MEPS full-year consolidated file (2022)
HC-251	MEPS full-year consolidated file (2023)
ED	Emergency Department visit
IP	Inpatient stay / hospital admission
SES	Socio-economic status
LONGWT	Longitudinal person weight
VARSTR	Variance estimation stratum (survey design)
VARPSU	Primary sampling unit (survey design)
AUC	Area Under the ROC Curve
PR-AUC	Precision–Recall AUC (Average Precision)
TP/TN/FP/FN	True Positive/True Negative/False Positive/False Negative
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
R^2	Coefficient of determination
RF	Random Forest
XGB	XGBoost
MLP	Multilayer Perceptron

B. Variable Glossary

VARIABLE	MEANING / DEFINITION	NOTES / DERIVATION
TOTEXPY1	Total health care expenditures, Year-1	MEPS total expenditure
TOTEXPY2	Total health care expenditures, Year-2	MEPS total expenditure
LOG_TOTEXPY1	log(1 + TOTEXPY1)	Engineered predictor
LOG_TOTEXPY2	log(1 + TOTEXPY2)	Regression target
HIGHCOST_Y2	High-cost indicator in Year-2	Top-cost percentile (as defined in thesis)
ERTOTY2	Total ED visits, Year-2	MEPS count variable
IPDISY2	Total inpatient discharges, Year-2	MEPS count variable
ANY_ED_Y1	Any ED visit in Year-1	Indicator: ERTOTY1 > 0
ANY_ED_Y2	Any ED visit in Year-2	Indicator: ERTOTY2 > 0
ANY_IP_Y1	Any inpatient stay in Year-1	Indicator: IPDISY1 > 0
ANY_IP_Y2	Any inpatient stay in Year-2	Indicator: IPDISY2 > 0
AGE_GROUP	Age group (0–17, 18–44, 45–64, 65+)	Engineered from AGEY1X/AGEY2X
SEX_BIN	Female indicator	1 if SEX = 2
RACE_ETH	Race/ethnicity label	Engineered from RACETHX
REGIONY1_CAT	Region label (Year-1)	Engineered from REGIONY1
EDU_GROUP	Education group	Engineered from EDUCYR
POVCATY1_CAT	Poverty category label	Engineered from POVCATY1
FAMSIZE_Y1	Family size (Year-1)	From FAMSZEY1
FAMSIZE_Y1_GRP	Family size group (1–6, 7+)	Engineered from FAMSZEY1
INS_TYPE_Y1	Insurance type (Year-1)	Engineered from INSURCY1
WORKED_Y1	Ever worked in Year-1	Engineered from EVRWRKY1
LOG_FAMINCY1	log(1 + family income, Y1)	Engineered from FAMINCY1
LOG_UNEMP_COMP_Y1	log(1 + unemployment compensation, Y1)	Engineered from UNEMPY1X
ANY_UNEMP_COMP_Y1	Any unemployment compensation, Y1	Indicator: UNEMPY1X > 0
RTHLTH1_FAIRPOOR	Fair/poor self-rated health, R1	Engineered from RTHLTH1

MNHLTH1_FAIRPOOR	Fair/poor mental health, R1	Engineered from MNHLTH1
HIBPDXY1	High blood pressure, Year-1	MEPS condition indicator
HIBPDXY1_BIN	Hypertension indicator	Engineered from HIBPDXY1
CHOLDXY1_BIN	High cholesterol indicator	Engineered from CHOLDXY1

Appendix E: Ethical Considerations

- **Intended use:** The models are intended for **risk stratification / top-k prioritization** under capacity constraints. Outputs support human review rather than automated decisions.
- **Non-use:** Not for diagnosis, not for causal claims, and not as the sole basis for denying services/coverage.
- **Potential harms:**
    - *False positives:* unnecessary outreach or stigma.
    - *False negatives:* missed high-risk individuals (especially low-prevalence outcomes such as inpatient).
    - *Mitigation:* use ranking/top-k with human oversight; avoid hard thresholds as the only decision rule; communicate uncertainty.
- **Fairness:** Model performance may differ across subgroups (e.g., age/sex). This thesis reports subgroup metrics (e.g., recall@top-k, PR-AUC) and recommends continuous monitoring and transparent reporting if used operationally.
- **Interpretability:** "Reason tags" in the MVP are **rule-based summaries** of baseline inputs to improve readability and plausibility checks. They are **not model-derived** and should not be interpreted as feature importance or causal drivers.

Appendix F. Data Use Compliance

This thesis uses the public-use MEPS-HC Panel 27 longitudinal file (HC-252). In this repository, we include derived, feature-engineered datasets (e.g., df_feat.parquet) and trained model artifacts for reproducibility. These derived outputs are created from public-use MEPS data and are therefore subject to the same public-use restrictions. In particular, no attempt is made to re-identify individuals, and the data are not linked to any individually identifiable external datasets. Results are reported only in aggregate form, and shared artifacts are intended for research and demonstration purposes only. Users of this repository are responsible for ensuring continued compliance with MEPS public-use requirements.

Appendix G. Reproducibility

All code used in this thesis is provided in the accompanying GitHub repository: < https://github.com/WenxiGu/Thesis-project-MEPS >. The analysis is organized into notebooks for EDA and modelling, and a src/ package for reusable preprocessing, feature engineering, and modelling utilities. A Streamlit MVP and trained model artifacts are included for demonstration.

Code structure

- notebooks/: end-to-end workflow (EDA → feature engineering → modelling → evaluation).
- src/: reusable data I/O, preprocessing, feature engineering, and modelling utilities.
- app/: Streamlit MVP for inference and top-k selection.

- results/: figures, tables, and exported model artifacts.

Random seeds

All models use a fixed random seed to ensure reproducibility:

- RANDOM_SEED = 42 (defined in config.py and reused across scripts).
- Model training scripts and notebooks pass random_state=42 where applicable.

Environment

- Python 3.12.
- Dependencies pinned in app/environment_app.yml (mvp), notebooks/environment_thesis.yml (modelling).

Re-running the analysis

1. Install dependencies (**conda** env create -f notebooks/environment_thesis.yml).
2. Run notebooks in order:
   01_EDA.ipynb → 02_Data Preparation & Feature Engineering.ipynb →
   03_Modelling_baseline.ipynb → 04_Model Predictions.ipynb → 05_weighted evaluation.ipynb.
3. (Optional) Retrain and export preferred models:
   train_and_save_preferred_models.py

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